Eddy-covariance data processing and analysis with *Reddy*

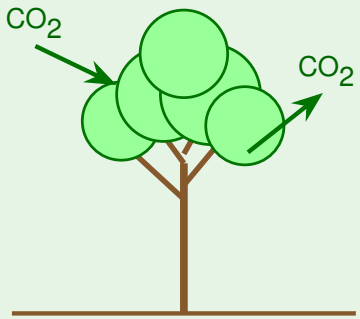
29.10.2026 | Laura Mack



Applications of eddy-covariance (EC)

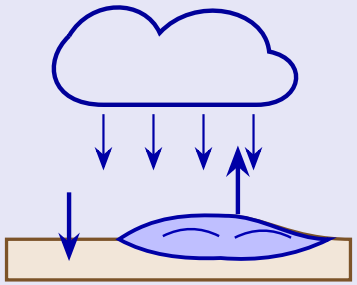
EC: continuous measurements of turbulent land–atmosphere exchanges

Biogeochemistry



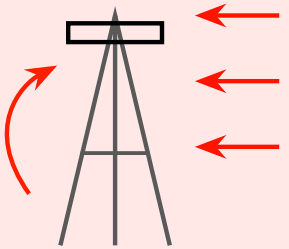
Carbon, water and energy exchange of ecosystems

Hydrology



Evapotranspiration and water vapor fluxes

Meteorology



Boundary-layer dynamics and turbulence

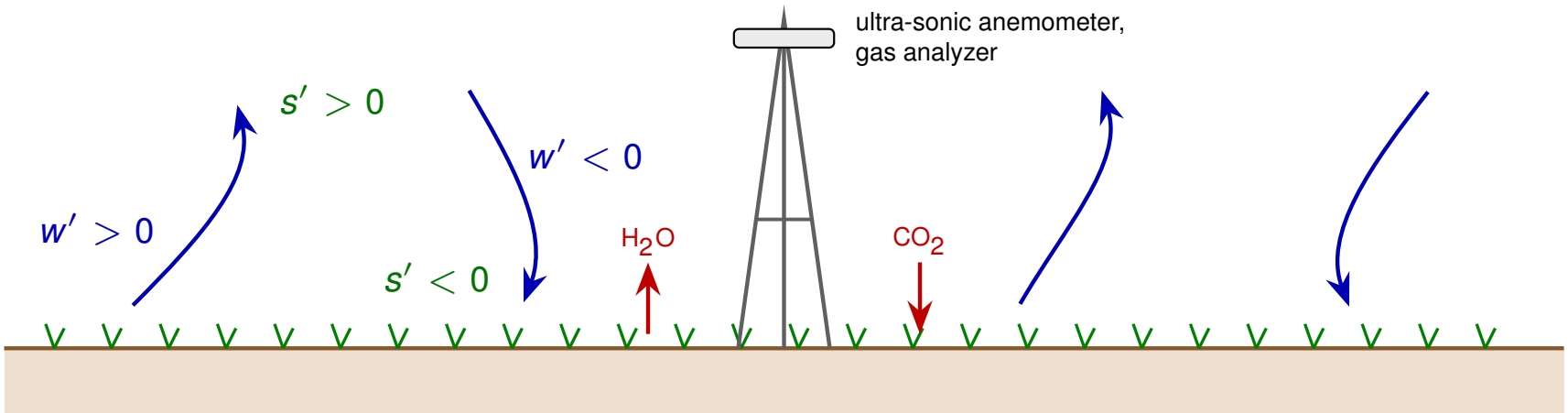
The eddy-covariance principle

EC principle: vertical kinematic flux of scalar s $F_s = \overline{w's'}$

- assumptions: stationarity, well-developed turbulence, interpretable source area, ...
  $\rightarrow$ challenging in heterogeneous and complex terrain

Fast measurements: 10–20 Hz
vertical wind speed w and scalar s

Turbulent flux
$F_s = \overline{w's'}$



Correlating simultaneous fluctuations in vertical wind speed and a scalar gives the net turbulent surface–atmosphere flux, typically averaged over 30 minutes.

Existing EC software

Raw-data processing:

- EddyPro (Fortran) (LI-COR Biosciences, 2017)
- TK3 (Fortran) (Mauder and Foken, 2011)
- EddyUH (Matlab) (Mammarella et al., 2016)
- eddy4R (R) (Metzger et al., 2017)

Post-processing:

- ONEFlux (C) (Pastorello et al., 2020)
- REddyProc (R) (Wutzler et al., 2018)
- openeddy (R) (McGloin et al., 2018)
- flux-data-qaqc (Python) (Volk et al., 2021)

Specific Applications

- bigleaf (R) (Knauer et al., 2018)
- FluxnetLSM (R) (Ukkola et al., 2017)
- ...

→ Reddy: modular, site-specific processing and advanced turbulence diagnostics

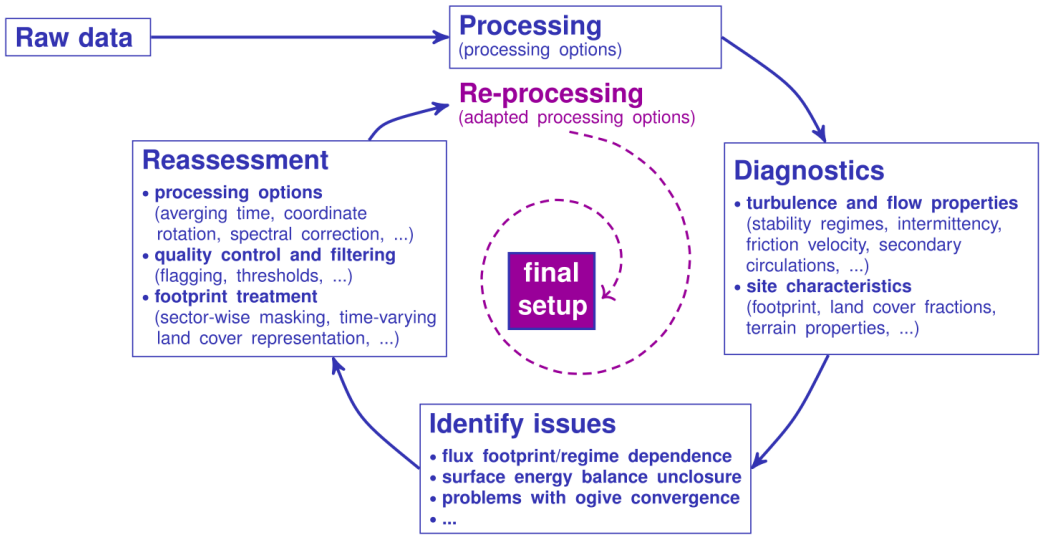
Reddy: Philosophy

Aim: Flexible toolkit for key eddy-covariance data analysis steps

Special features of Reddy:

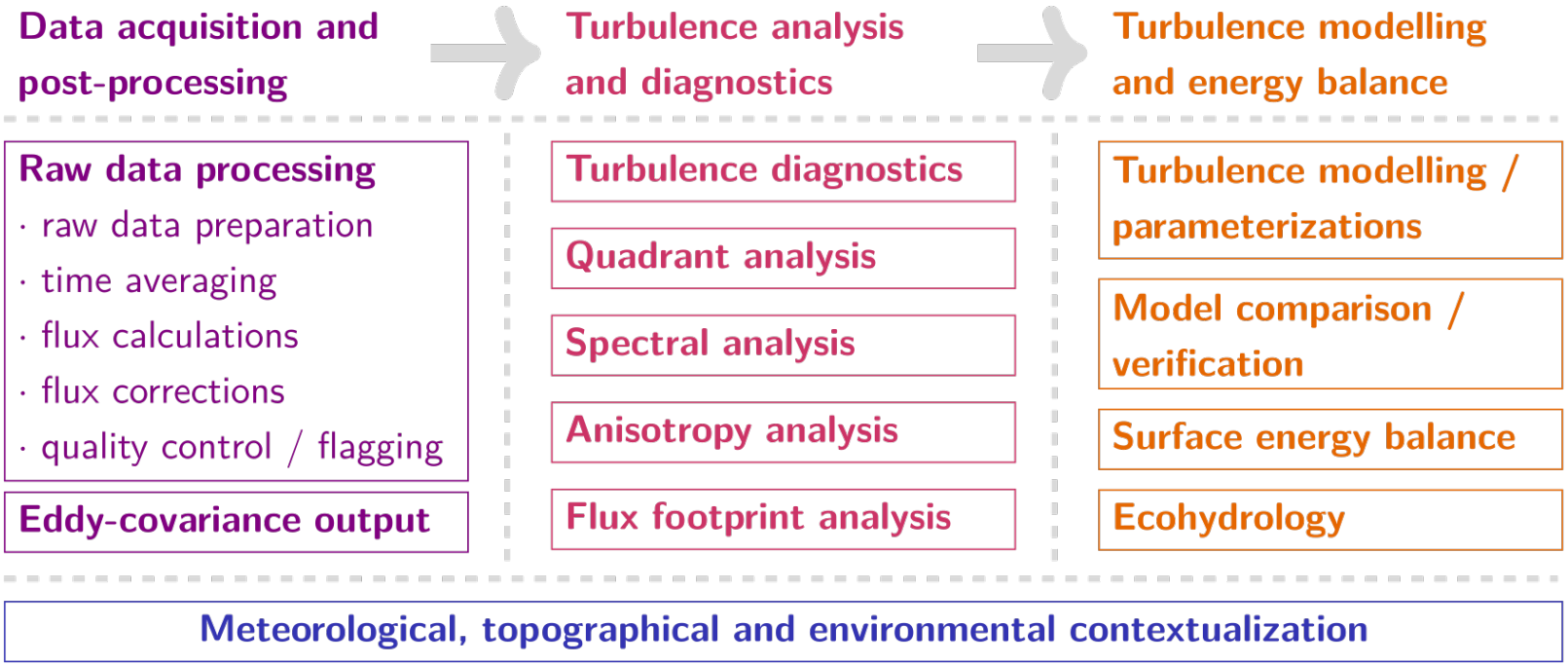
- fully modular design
- extended functionality (e.g. anisotropy, ogives, ...)
- tools to work with LES and NWP model output

Re-evaluation spiral for eddy-covariance data processing in 'Reddy'



→ **applications**: support research and flux station monitoring

Reddy: Modular structure

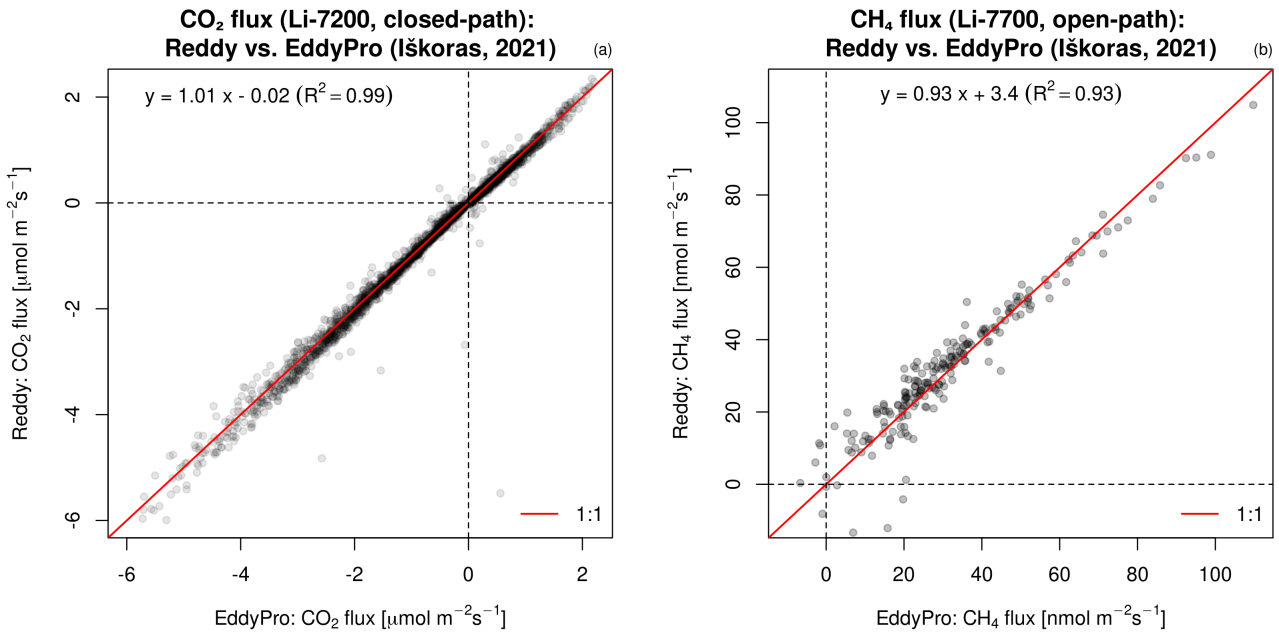


Reddy: EC processing

Raw-data processing in Reddy: separate functions → customized processing pipeline

Step 1: Raw data preparation
Step 2: Tilt correction, time averaging and flux calculation
Step 3: Flux corrections
Step 4: Quality control and flagging

Comparison of flux estimates between Reddy and EddyPro



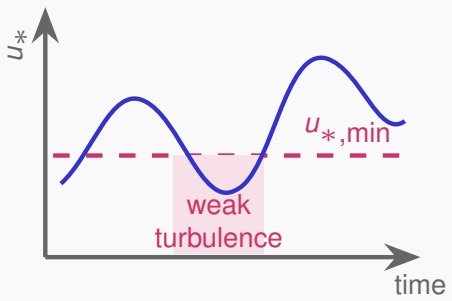
Reddy: Turbulence diagnostics

Calculation of turbulence diagnostics

- friction velocity, Obukhov length, stability parameter, decoupling metric, ...

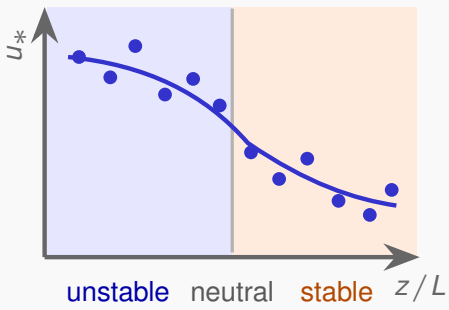
  → joint use of several diagnostics to assess flow properties

Friction velocity



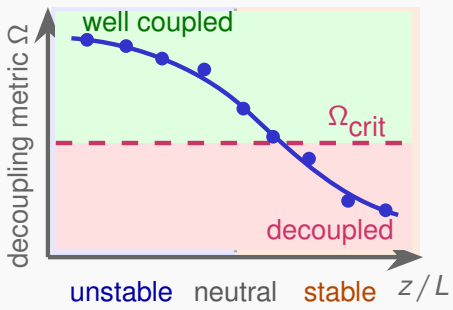
u_* indicates the efficiency of turbulent momentum transfer.

Stability parameter



Stable conditions often coincide with weaker and intermittent turbulence.

Surface (de)coupling



Low Ω under stable conditions indicates reduced surface–atmosphere coupling.

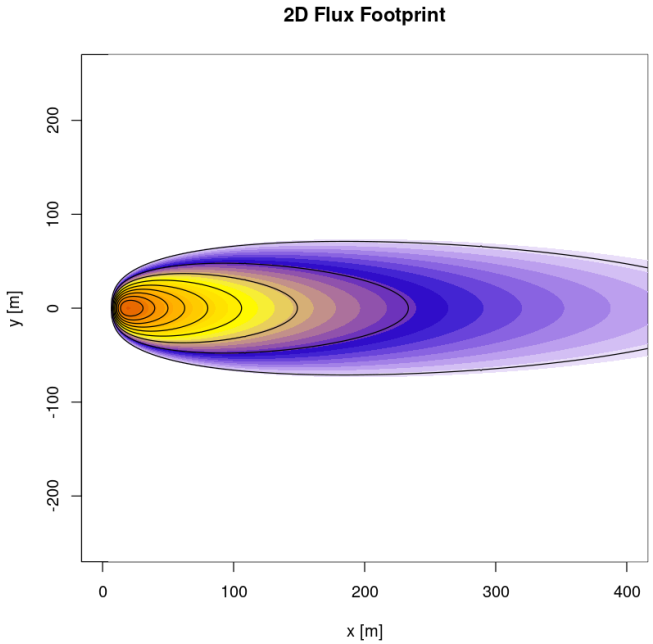
Reddy: Flux footprint estimation

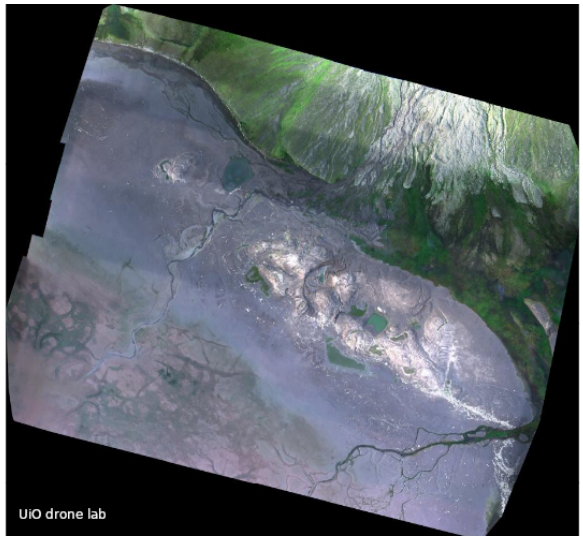
Flux footprint: Upwind surface area that contributes to the measured EC flux

Flux footprint calculation in Reddy:

- available models: (Kormann and Meixner, 2001) and (Kljun et al., 2015)
- flux footprint climatology, geo-referencing

→ assumptions: homogeneous surface, roughness length, displacement height





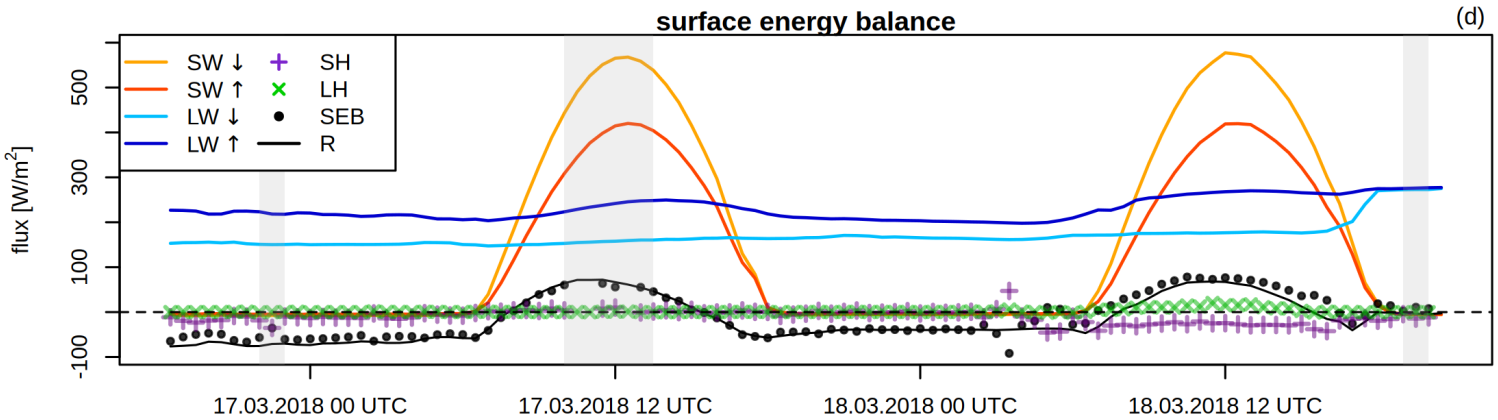
Use case: combine footprint climatology with drone-based vegetation classification to quantify contributing land-cover types.

Reddy: Surface energy balance

Surface energy balance: $SW\downarrow - SW\uparrow + LW\downarrow - LW\uparrow - G - S = SH + LH$

- ancillary measurement: radiation, soil heat flux / storage terms, ...

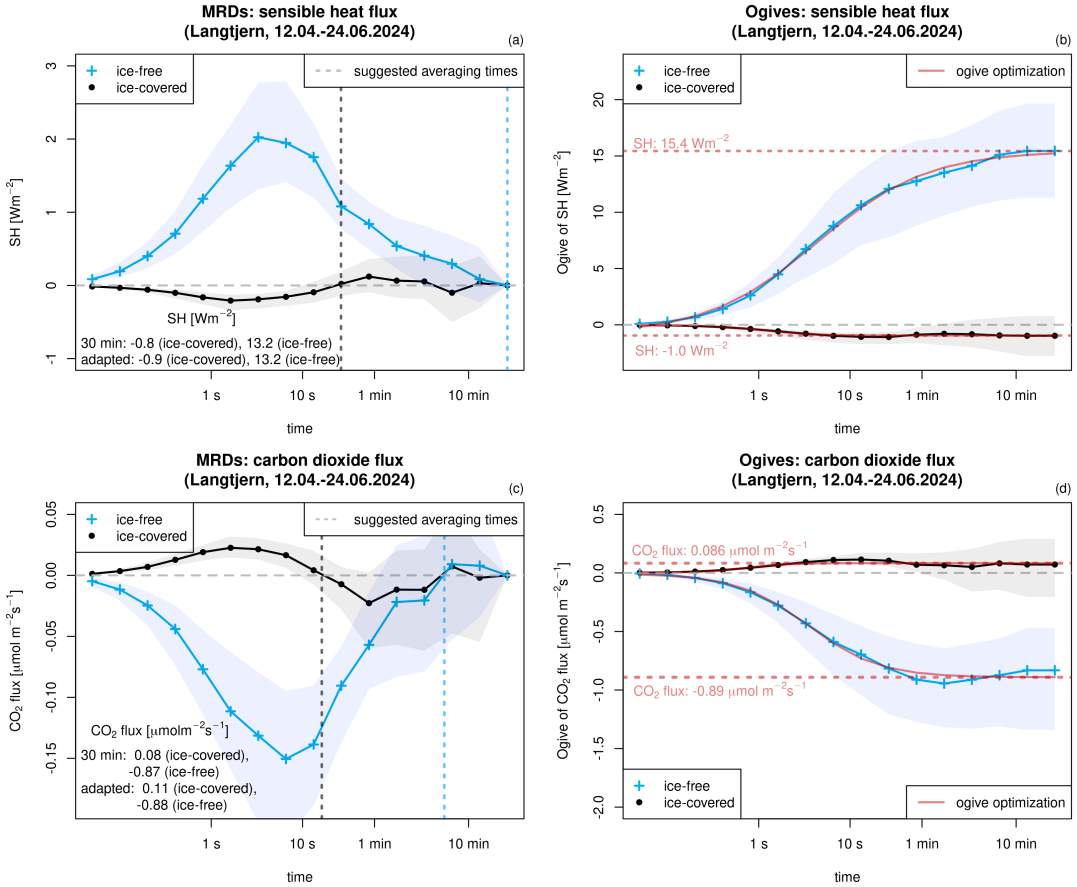
  → heterogeneous sites have commonly larger imbalance



Reddy: Spectra and Ogives

How to select an appropriate averaging time for the flux calculation?

- multi-resolution decomposition (MRD): identifies timescales contributing to covariance (turbulence vs. low-frequency contributions)
- ogive optimization: estimates flux from ogive convergence (without prescribed one fixed averaging time)

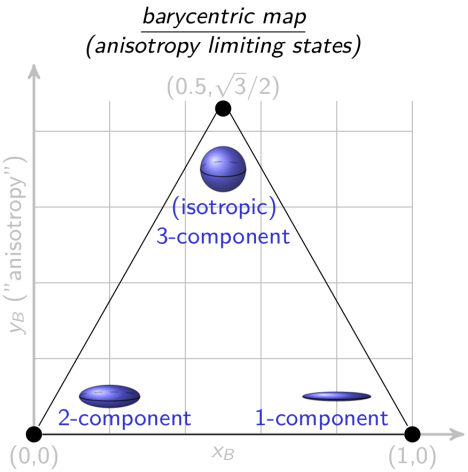


→ assess low-frequency motions and non-stationarity

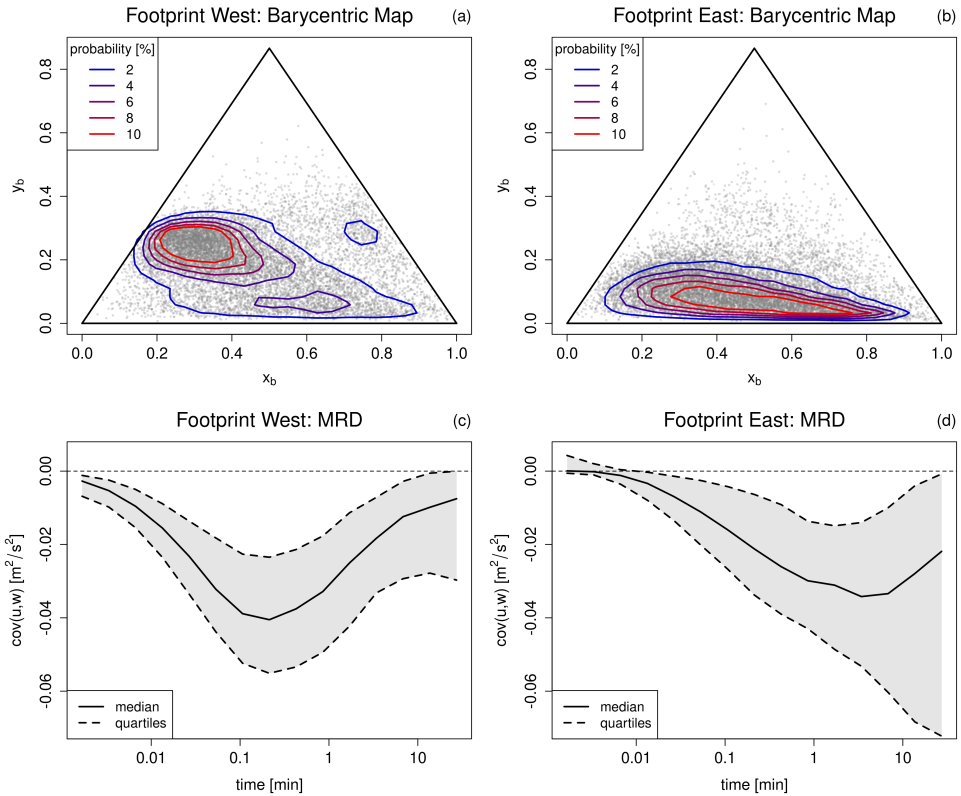
Reddy: Anisotropy analysis

How does the geometry of turbulence differ under different flow conditions?

- anisotropy analysis:
  invariant analysis of Reynolds stress tensor
- barycentric map:
  visualization of anisotropy limiting states



→ assess turbulence geometry

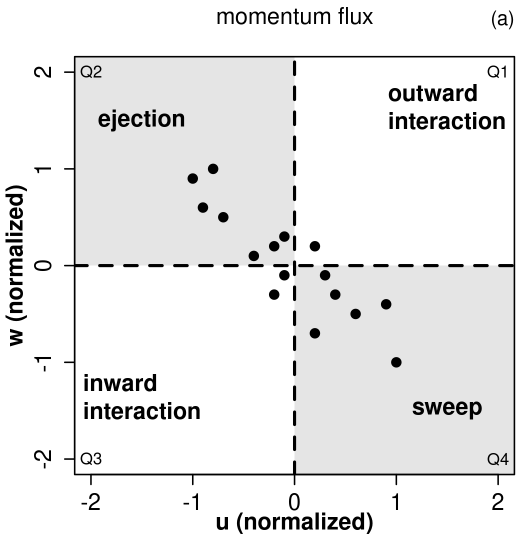


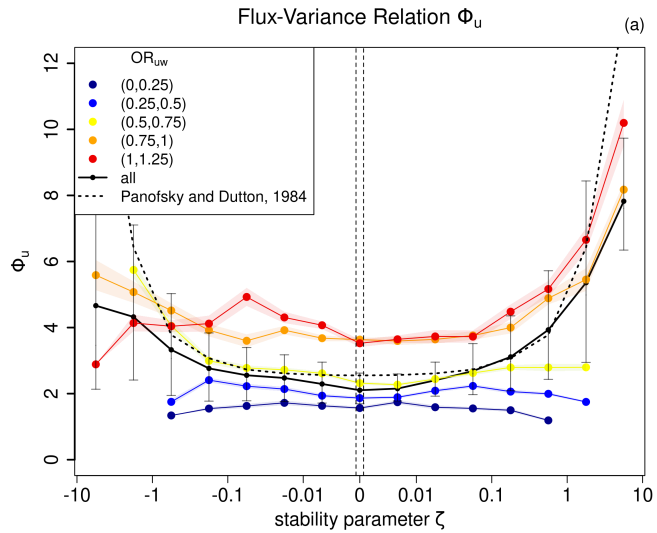
Figures: Barycentric map and MRD cospectra for footprint west and east @Finse, Norway.

Reddy: Further Tools

Further micro-meteorological tools available in Reddy:

- quadrant analysis
- flux-variance relations
- structure functions
- flux partitioning
- auxiliary: background meteorology, topography characterization



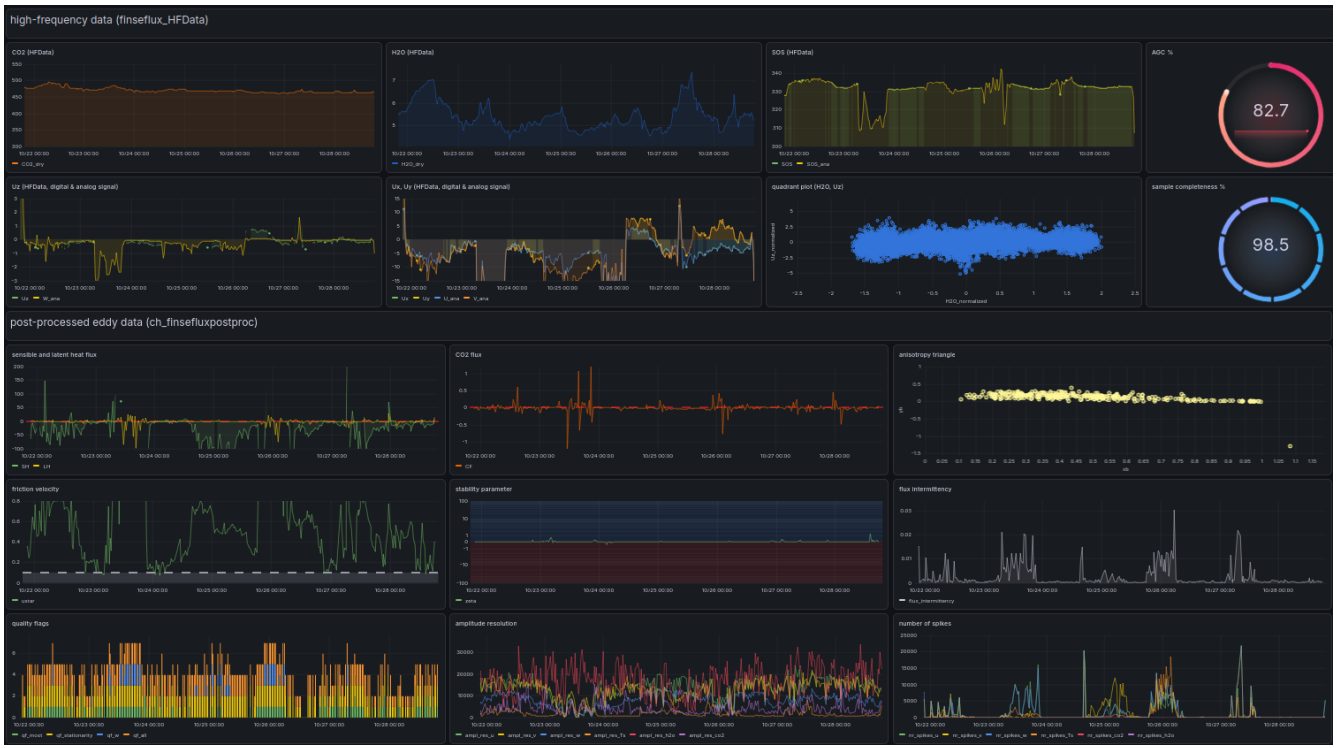


→ assess coherent structures, transfer efficiencies etc. depending on flow conditions

Reddy: Application areas

Applications:

- support research
- station monitoring



Reddy: Summary and Links

Specific features of Reddy:

- fully modular design
- extended functionality (e.g. anisotropy analysis, ogive optimization)
- tools to work with LES and NWP model output

→ feedback, suggestions, contributions welcome :)

Links:

GitHub repository: https://github.com/noctiluc3nt/Reddy

GitHub page (tutorials, jupyter notebooks): https://noctiluc3nt.github.io/ec_analyze/

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(GitHub repo)

(demo)

(article)